

Horizontal Cabling Standards

Permanent Link Limits

The horizontal permanent link is limited to 90 metres of solid-conductor cabling. With patch and equipment cords, the total channel length must not exceed 100 metres. These limits apply to all balanced twisted-pair categories from Cat 5e through Cat 8.

- Permanent link: 90 m maximum
- Channel: 100 m maximum
- Patch cords: 5 m at each end recommended

Category Performance

Cat 6 supports 1 Gbps to 100 metres. Cat 6A supports 10GBASE-T to 100 metres, subject to the 90-metre permanent-link limit and alien-crosstalk mitigation. Cat 8 supports 25/40 Gbps to 30 metres in the data centre.

Work Area Components

The work area consists of the wall outlet or floor box, the patch cord to the end device, and any transition hardware. TIA-568-D requires a minimum of two outlets per work area.